

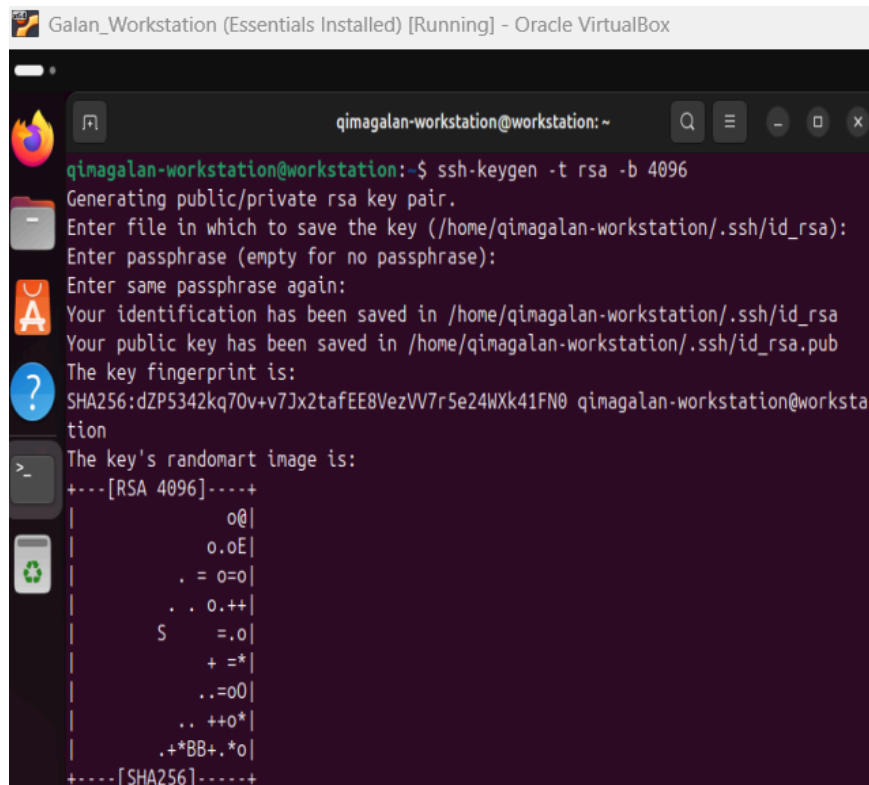
COURSE CODE/TITLE:	CPE 212 / Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	7/16/2026	DATE SUBMITTED:	7/23/2026
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ACTIVITY NO. 2 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

- The simplest way to generate a key pair is to run *ssh-keygen* without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users *.ssh* directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case *id_rsa* when using the default RSA algorithm. It could also be, for example, *id_dsa* or *id_ecdsa*.
- Issue the command *ssh-keygen -t rsa -b 4096*. The algorithm is selected using the *-t* option and key size using the *-b* option.



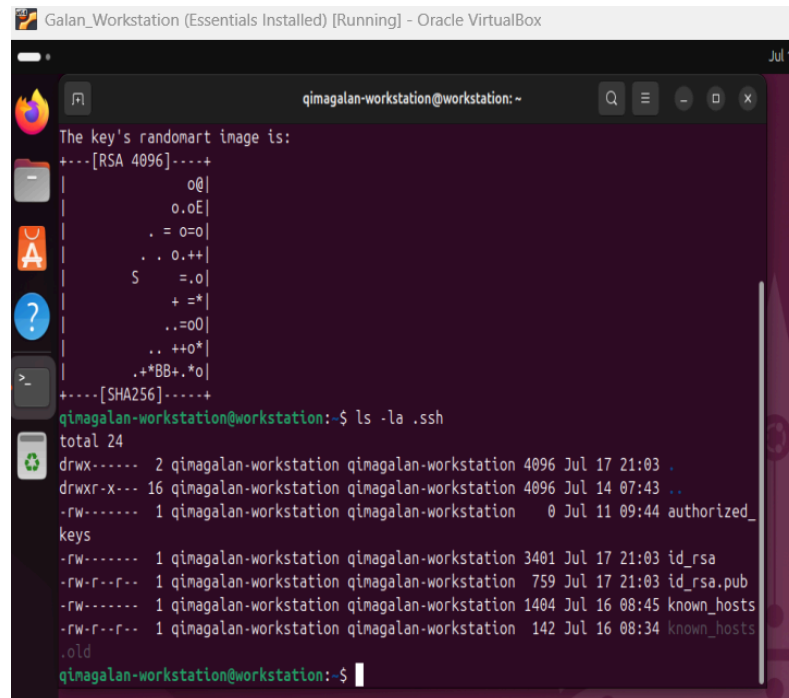
```

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox

qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/qimagalan-workstation/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/qimagalan-workstation/.ssh/id_rsa
Your public key has been saved in /home/qimagalan-workstation/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:dZP5342kq70v+v7Jx2tafEE8VezVV7r5e24Wxk41FN0 qimagalan-workstation@worksta
tion
The key's randomart image is:
+----[RSA 4096]-----+
|
| o@|
| o.oE|
| . = o=o|
| . . o.++|
| S   =.o|
| + =*|
| ..=o0|
| .. ++o*|
| .+BB+. *o|
+-----[SHA256]-----+

```

3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.
4. Verify that you have created the key by issuing the command `ls -la .ssh`. The command should show the .ssh directory containing a pair of keys. For example, `id_rsa.pub` and `id_rsa`.



```

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~
The key's randomart image is:
+---[RSA 4096]---+
|                o@|
|               o.oE|
|              . = 0=0|
|             . . 0.++|
|            S   =.o|
|           +  =*|
|          ..=o0|
|         .. ++o*|
|        .+*BB+.o|
+---[SHA256]-----+
qimagalan-workstation@workstation:~$ ls -la .ssh
total 24
drwx----- 2 qimagalan-workstation qimagalan-workstation 4096 Jul 17 21:03 .
drwxr-x--- 16 qimagalan-workstation qimagalan-workstation 4096 Jul 14 07:43 ..
-rw----- 1 qimagalan-workstation qimagalan-workstation  0 Jul 11 09:44 authorized_
keys
-rw----- 1 qimagalan-workstation qimagalan-workstation 3401 Jul 17 21:03 id_rsa
-rw-r--r-- 1 qimagalan-workstation qimagalan-workstation  759 Jul 17 21:03 id_rsa.pub
-rw----- 1 qimagalan-workstation qimagalan-workstation 1404 Jul 16 08:45 known_hosts
-rw-r--r-- 1 qimagalan-workstation qimagalan-workstation 142 Jul 16 08:34 known_hosts
old
qimagalan-workstation@workstation:~$

```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an *authorized_keys* file. This can be conveniently done using the *ssh-copy-id* tool.
2. Issue the command similar to this: *ssh-copy-id -i ~/.ssh/id_rsa user@host*

Server 1 Key

```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa qimagalan-workstation@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/qimagalan-workstation/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
qimagalan-workstation@server1's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'qimagalan-workstation@server1'"
and check to make sure that only the key(s) you wanted were added.
qimagalan-workstation@workstation:~$
```

Server 2 Key

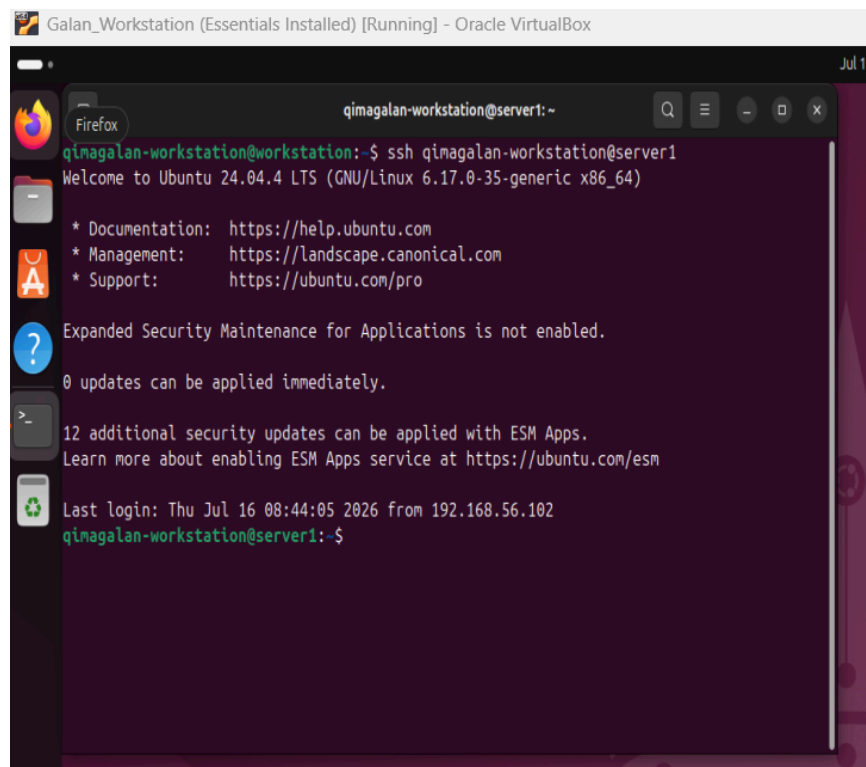
```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ ssh-copy-id -i ~/.ssh/id_rsa qimagalan-workstation@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/qimagalan-workstation/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
qimagalan-workstation@server2's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'qimagalan-workstation@server2'"
and check to make sure that only the key(s) you wanted were added.
qimagalan-workstation@workstation:~$
```

3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.
4. On the local machine, verify that you can SSH with Server 1 and Server 2.

What did you notice? Did the connection ask for a password? If not, why?

I noticed that I was able to automatically connect to the remote machine. This happens because we did not set any password to the key we created and therefore we automatically connect with the ssh key alone.

Server1



```

Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@server1:~
qimagalan-workstation@workstation: $ ssh qimagalan-workstation@server1
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

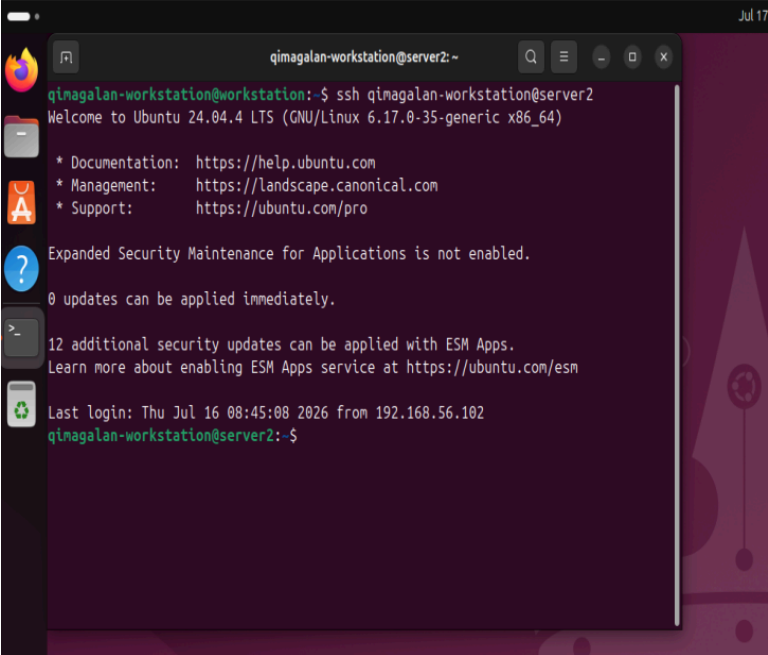
Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

12 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jul 16 08:44:05 2026 from 192.168.56.102
qimagalan-workstation@server1:~$
  
```

Server2

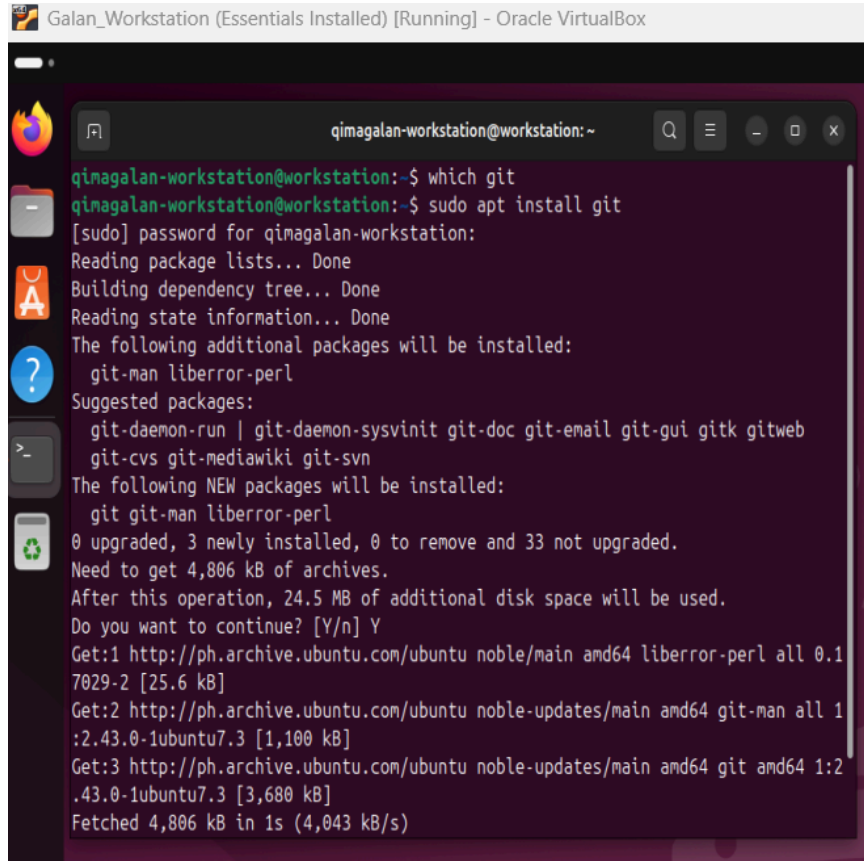


The screenshot shows a terminal window titled "qimagalan-workstation@server2: ~". The terminal displays the output of an SSH command executed from a workstation. The output includes a welcome message for Ubuntu 24.04.4 LTS, system information, links to documentation, management, and support, a security notice about ESM, and the last login time.

```
qimagalan-workstation@server2:~  
qimagalan-workstation@workstation:~$ ssh qimagalan-workstation@server2  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-35-generic x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
? Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
  
> 12 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
🗑 Last login: Thu Jul 16 08:45:08 2026 from 192.168.56.102  
qimagalan-workstation@server2:~$
```

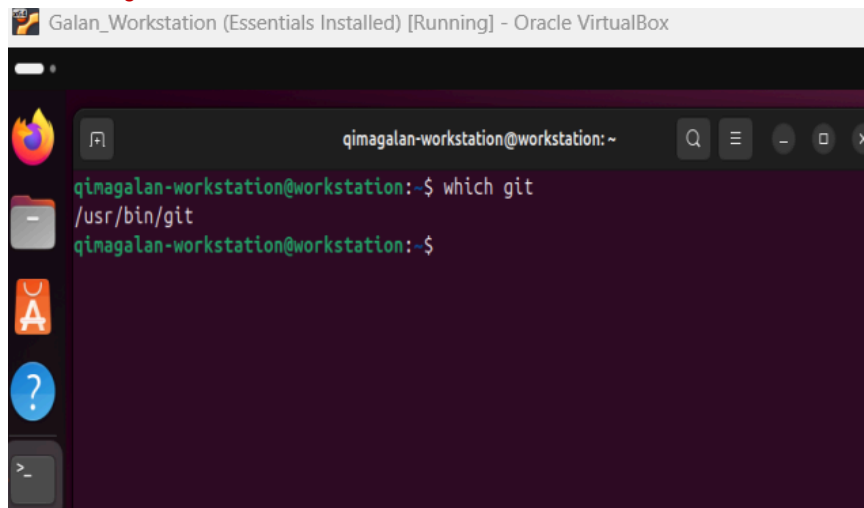
Task 3: Set up the Git Repository

1. On the local machine, verify the version of your git using the command *which git*. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: *sudo apt install git*



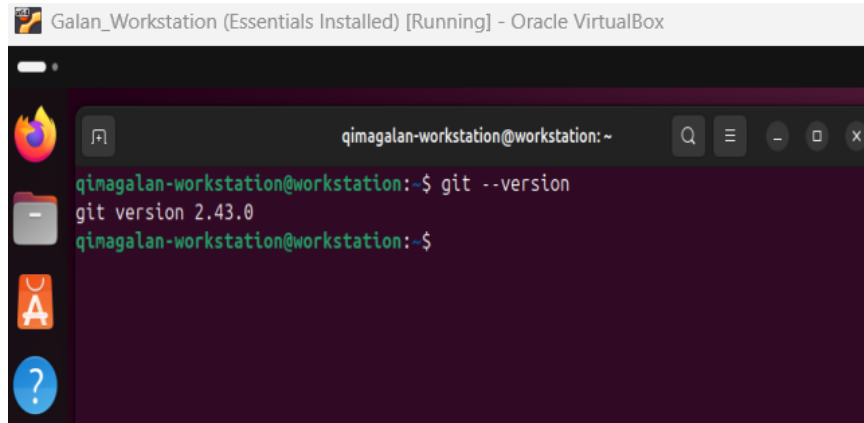
```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ which git
qimagalan-workstation@workstation:~$ sudo apt install git
[sudo] password for qimagalan-workstation:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 33 not upgraded.
Need to get 4,806 kB of archives.
After this operation, 24.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://ph.archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.1
7029-2 [25.6 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1
:2.43.0-1ubuntu7.3 [1,100 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2
.43.0-1ubuntu7.3 [3,680 kB]
Fetched 4,806 kB in 1s (4,043 kB/s)
```

2. After the installation, issue the command *which git* again. The directory of git is usually installed in this location: */usr/bin/git*.



```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~
qimagalan-workstation@workstation:~$ which git
/usr/bin/git
qimagalan-workstation@workstation:~$
```

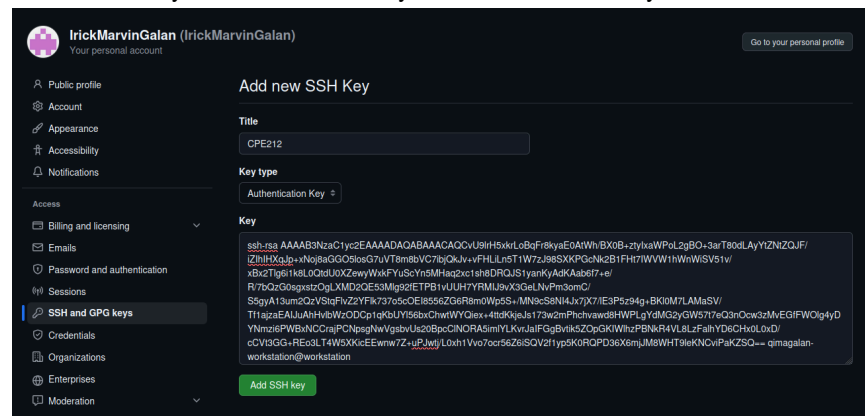
- The version of git installed in your device is the latest. Try issuing the command `git --version` to know the version installed.



```
Galan_Workstation (Essentials Installed) [Running] - Oracle VM VirtualBox

qimagalan-workstation@workstation:~$ git --version
git version 2.43.0
qimagalan-workstation@workstation:~$
```

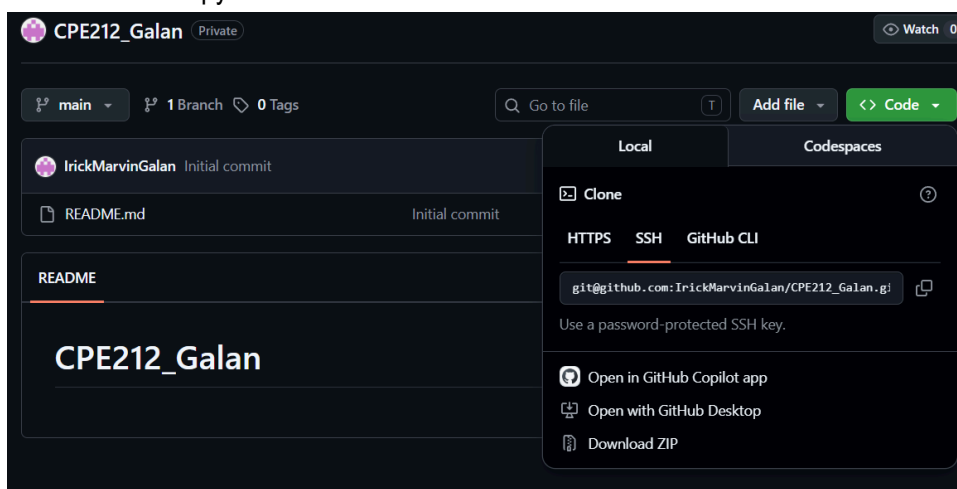
- Using the browser in the local machine, go to www.github.com.
- Sign up in case you don't have an account yet. Otherwise, login to your GitHub account.
 - Create a new repository and name it as CPE212_yourname. Check Add a README file and click Create repository.
 - Create a new SSH key on GitHub. Go to your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.



- c. On the local machine's terminal, issue the command `cat .ssh/id_rsa.pub` and copy the public key. Paste it on the GitHub key and press Add SSH key.

```
qimagalan-workstation@workstation: ~  
qimagalan-workstation@workstation:~$ cat .ssh/id_rsa.pub  
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCvU9lrH5xkrLoBqFr8kyaE0AtWh/BX0B+ztyIxaWPo  
L2gB0+3arT80dLAyYtZntZQJF/iZIHIXqJp+xNoj8aGG05losG7uVT8m8bVC7ibjQkJv+vFHLiLn5T1  
W7zJ98SXKPGcNk2B1Fht7IWVW1hWnWiSV51v/xBx2Tlg6i1k8L0Qtdu0XZewyWxkFYuScYn5MHaq2xc1  
sh8DRQJS1yanKyAdKAab6f7+e/R/7bQzG0sgxstz0gLXMD2QE53Mlg92fETPB1vUUH7YRMIJ9vX3GeLN  
vPm3omC/S5gyA13um2QzVStqFlvZ2YFlk737o5cOEI8556ZG6R8m0Wp5S+/MN9cS8NI4Jx7jX7/LE3P5  
z94g+BKL0M7LAMA5V/Tf1a1zaEAIJuAhHv1bWz0DCp1qKbUYL56bxChwtWYQieX+4ttDKkJeJs173w2m  
Phchvawd8HWPLgYdMG2yGW57t7eQ3n0cw3zMvEGfFW0lg4yDYNmzi6PBWxNCCraJPCNpsgNwVgsbvUs2  
0BpcCLNORA5imLYLKvrJaIFGgBvtik5Z0pGKIWlhZPBnkR4VL8LzFalHYD6CHx0L0xD/cCVt3GG+REo3  
LT4W5XKicEEwnw7Z+uPJWtj/L0xh1Vvo7ocr56Z6iSQV2f1yp5K0RQPD36X6mJm8WHT9leKNCviPaKZ  
SQ== qimagalan-workstation@workstation  
qimagalan-workstation@workstation:~$
```

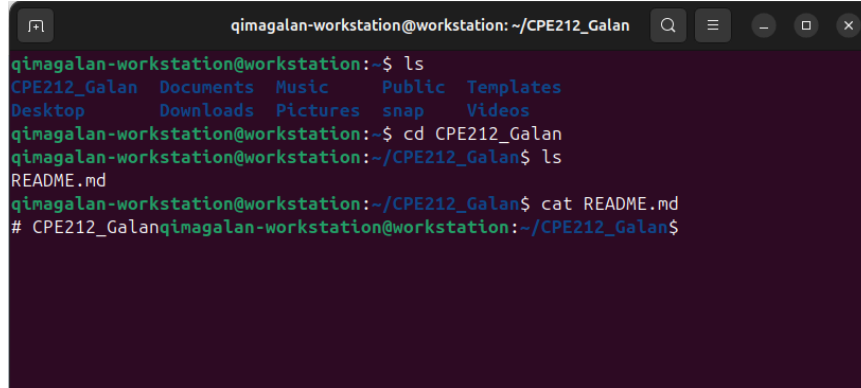
- d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.



- e. Issue the command `git clone` followed by the copied link. For example, `git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git`. When prompted to continue connecting, type yes and press enter.

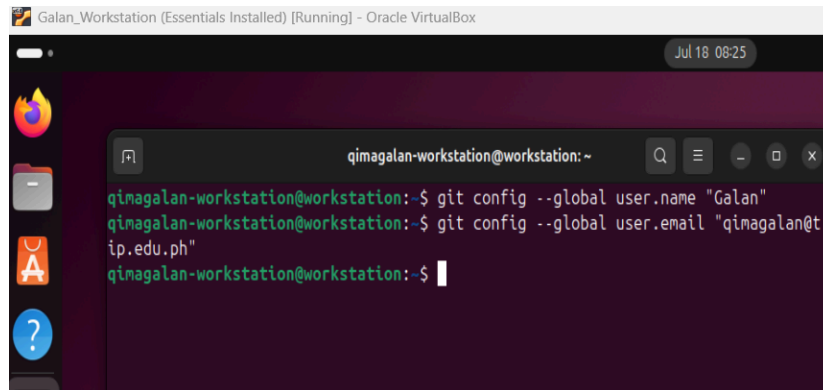
```
qimagalan-workstation@workstation: ~  
qimagalan-workstation@workstation:~$ git clone git@github.com:IrickMarvinGalan/CPE212_Galan.git  
Cloning into 'CPE212_Galan'...  
The authenticity of host 'github.com (20.205.243.166)' can't be established.  
ED25519 key fingerprint is SHA256:+DiY3wvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.  
remote: Enumerating objects: 3, done.  
remote: Counting objects: 100% (3/3), done.  
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
Receiving objects: 100% (3/3), done.  
qimagalan-workstation@workstation:~$
```

- f. To verify that you have cloned the GitHub repository, issue the command `ls`. Observe that you have the `CPE232_yourname` in the list of your directories. Use `CD` command to go to that directory and `LS` command to see the file `README.md`.

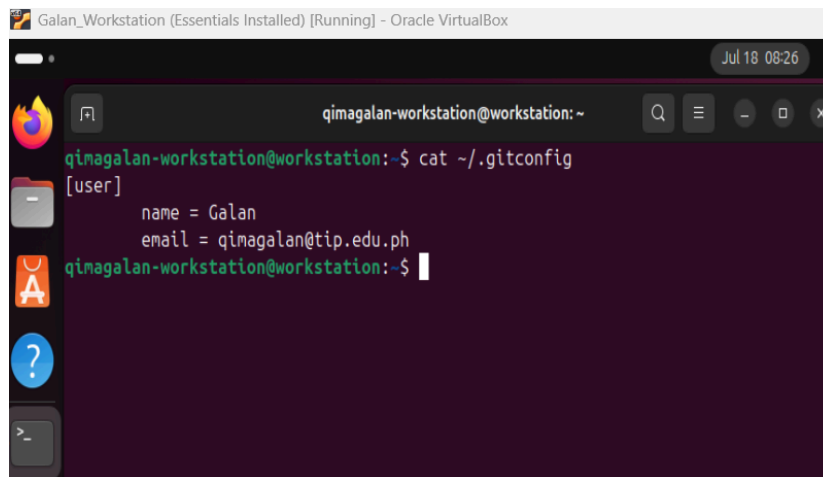


```
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~$ ls
CPE212_Galan  Documents  Music      Public  Templates
Desktop      Downloads  Pictures   snap    Videos
qimagalan-workstation@workstation:~$ cd CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ ls
README.md
qimagalan-workstation@workstation:~/CPE212_Galan$ cat README.md
# CPE212_Galanqimagalan-workstation@workstation:~/CPE212_Galan$
```

- g. Use the following commands to personalize your git.
- `git config --global user.name "Your Name"`
 - `git config --global user.email youname@email.com`
 - Verify that you have personalized the config file using the command `cat ~/.gitconfig`

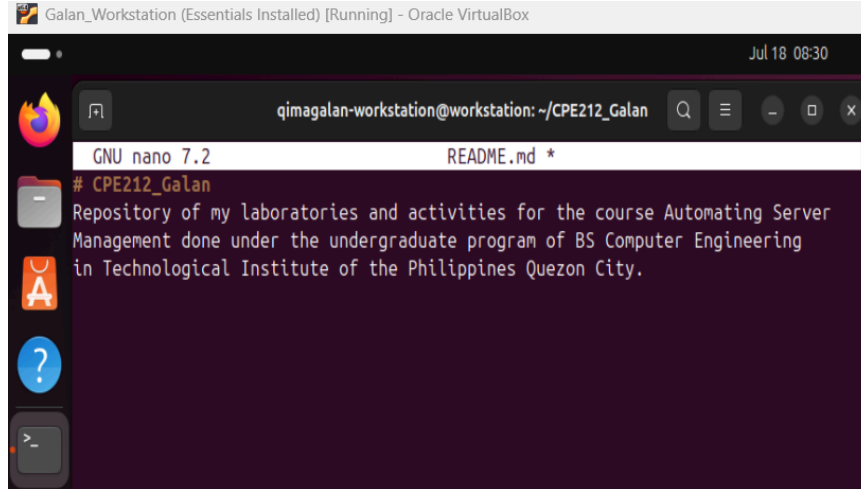


```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:25
qimagalan-workstation@workstation:~
qimagalan-workstation@workstation:~$ git config --global user.name "Galan"
qimagalan-workstation@workstation:~$ git config --global user.email "qimagalan@tip.edu.ph"
qimagalan-workstation@workstation:~$
```



```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:26
qimagalan-workstation@workstation:~
qimagalan-workstation@workstation:~$ cat ~/.gitconfig
[user]
    name = Galan
    email = qimagalan@tip.edu.ph
qimagalan-workstation@workstation:~$
```

- h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

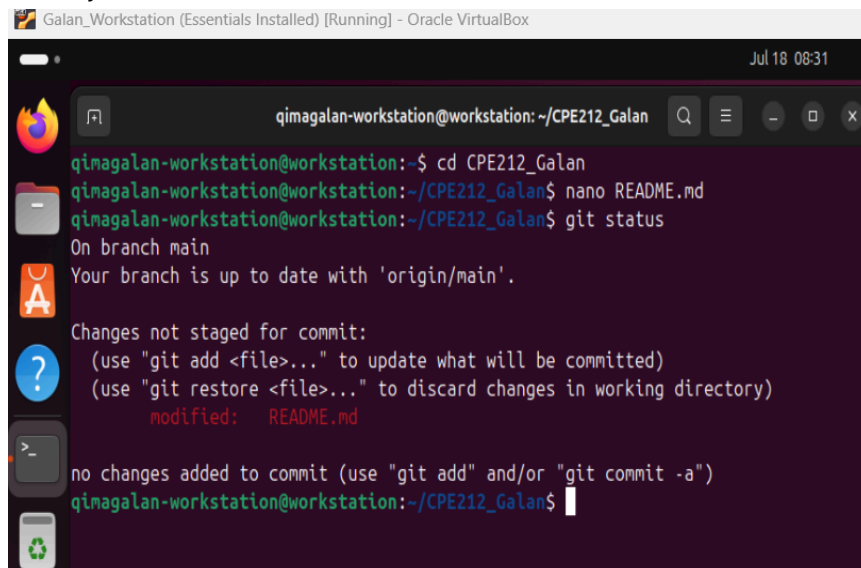


```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:30
qimagalan-workstation@workstation: ~/CPE212_Galan
GNU nano 7.2 README.md *
# CPE212_Galan
Repository of my laboratories and activities for the course Automating Server
Management done under the undergraduate program of BS Computer Engineering
in Technological Institute of the Philippines Quezon City.
```

- i. Use the `git status` command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven't, and which files aren't being tracked by Git. Status output does not show any information regarding the committed project history.

What is the result of issuing this command?

As we have discussed, git is a version control system, and we would see this also in action in terms of git status. The git detects that there is a mismatch in the content of the README.md from the snapshot of origin/main and the current version found within the local system.

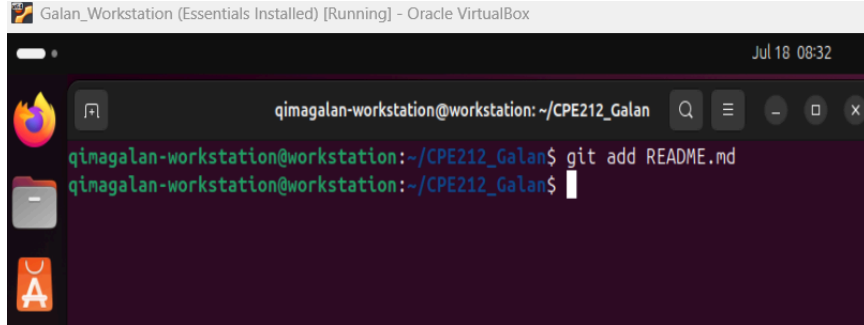


```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:31
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~$ cd CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ nano README.md
qimagalan-workstation@workstation:~/CPE212_Galan$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

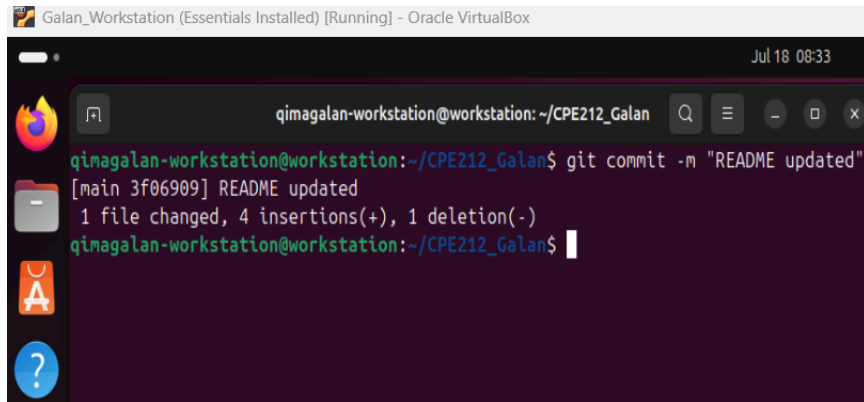
no changes added to commit (use "git add" and/or "git commit -a")
qimagalan-workstation@workstation:~/CPE212_Galan$
```

- j. Use the command *git add README.md* to add the file into the staging area.



```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:32
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ git add README.md
qimagalan-workstation@workstation:~/CPE212_Galan$
```

- k. Use the *git commit -m "your message"* to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next commit.

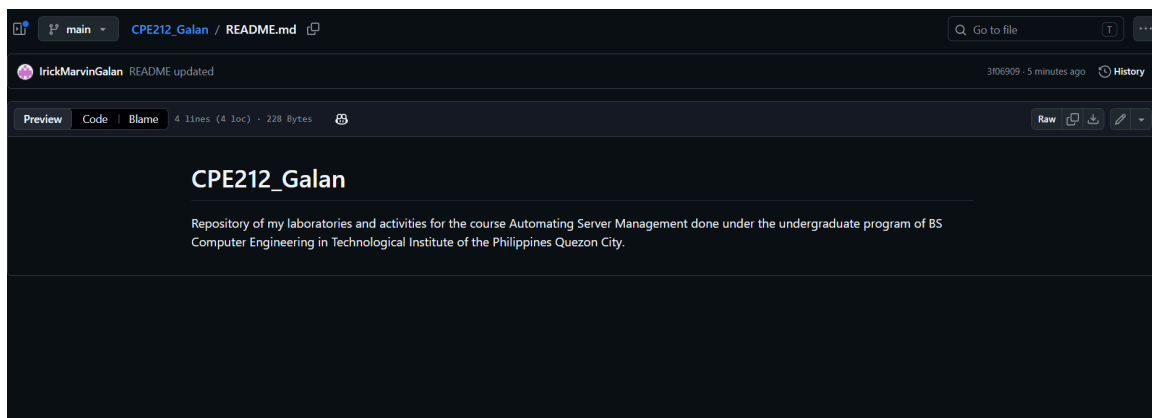


```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
Jul 18 08:33
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ git commit -m "README updated"
[main 3f06909] README updated
1 file changed, 4 insertions(+), 1 deletion(-)
qimagalan-workstation@workstation:~/CPE212_Galan$
```

- l. Use the command *git push <remote><branch>* to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue *git push origin main*.

```
Galan_Workstation (Essentials Installed) [Running] - Oracle VirtualBox
qimagalan-workstation@workstation: ~/CPE212_Galan
qimagalan-workstation@workstation:~/CPE212_Galan$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 3 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 417 bytes | 417.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:IrickMarvinGalan/CPE212_Galan.git
d664609..3f06909  main -> main
qimagalan-workstation@workstation:~/CPE212_Galan$
```

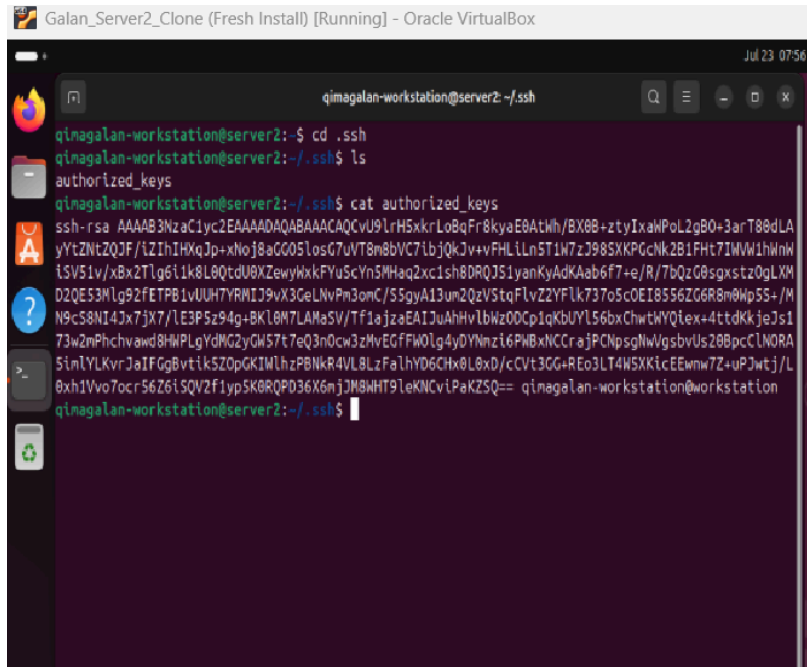
- m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

- How will you describe the ssh-program? What does it do?
The SSH program, if we try to understand its name "Secure Shell", the primary function of it is to allow secured remote connections. 'Shell' since ssh allows connection to the shell of the remote server via encrypted channel. This prevents other machines watching the communication channel from cracking the messages sent such as critical passwords.
- How do you know that you already installed the public key to the remote servers?



```
Galan_Server2_Clone (Fresh Install) [Running] - Oracle VirtualBox
Jul 23 07:56
qimagalan-workstation@server2: ~/.ssh
qimagalan-workstation@server2:~/.ssh$ cd .ssh
qimagalan-workstation@server2:~/.ssh$ ls
authorized_keys
qimagalan-workstation@server2:~/.ssh$ cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQCvU9LrH5xkrLoBqFr8kyaE8AtWh/BX8B+ztyIxaHPL2gB0+3arT80dLA
yYtZnt2QJF/iZIHxqJp+xNoj8aG005LosG7UVT8m8bVC7lbjQkQv+vFHLlLnST1W7zJ98SXKPGcNK2B1Fht7IMWW1hWnW
iSV51v/xBx2Tlg61k8L0QtdU8XZewyWxkFYu5cYn3MHaq2xc1sh8DRQJ51yanKyAdKAab6f7+e/R/7bQzC8sgxstz0gLXM
D2QE53Mlg92FETPB1VUUH7YRMI39vX3GeLNVm3omC/SSgyA13um2QzVStqFLvZ2YfLk737o5c0EI8556ZG6R8m0Wp5S+/M
N9c58NI4Jx7jX7/LE3P5z94g+BK18M7LAMA5V/Tf1ajzaEAIJuaHhV1bWz0DCp1qKbUYL56bxChwtHYQIex+4tttdKkjaJs1
73w2nPhchvawd8HNPgYdNG2yGMS7t7eQ3n0cw3zHvEGFW0Lg4yDYnmzi6PbXNCCraJPCNpsgNwVgsbvUs28BpcCUNORA
5inLYLkvrJaIFGgBvtik5Z0pGKIMhzPBnKR4VL8LzFalHYD6CHx8L8xD/cCVt3GG+Re03LT4N5XKicEEwmw7Z+uPjWtj/L
0xh1Vvo7ocr56Z6LSQV2f1yp5K0RQPD36X6njJN8HNT91eKNCvLPaKZ5Q== qimagalan-workstation@workstation
qimagalan-workstation@server2:~/.ssh$
```

By viewing the `.ssh/authorized_keys` file the key should be added, as we would see the key the authorized key generated by `ssh-keygen` could be found inside. And we would also see the label, `qimagalan-workstation@workstation` meaning that it came from the workstation machine.

3. What sort of things have we so far done to the remote servers using ansible commands?
From this laboratory activity we have not used any ansible commands yet.
4. How important is the inventory file?

Based on today's discussion, the inventory file contains the machines to manage, and it allows a single command to be reflected across the machines configured in the inventory file that makes systems administration scalable for a large company.

Conclusions

In this laboratory activity, I learned about how primarily ssh keys work. As discussed above, when generating a key there is always a private and public key pair. The public key is copied/installed on the remote machines so that the workstation having the private key could connect to those remote machines. A password could also be added to the key for increased security. I also noticed that github uses the same principle wherein I copied the public key of my workstation to github so that my machine could access it through the CLI and issue git commands remotely. I also learned here that git management has a basic core setup wherein it begins with cloning the repository, setting up the name and email in the CLI wherein the git connects to, modifications are then added to the stage and committed with a message. Finally the

push command allows the committed changes to be uploaded to github via the CLI, and it extracts the message along with it.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

No AIs were used for this laboratory activity

Purpose of Use:

N/A

Extent of Use:

N/A

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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